

Deribit Options Strategies

Investor Brief: Mechanics, Risks, and Return Profiles

Automated strategy engine on BTC and ETH options (Deribit). For discussion only; not investment advice.

This brief summarizes three configurable strategies implemented in the Deribit Options Strategy Engine: **naked short**, **bull put spread**, and **covered call**. The engine scans linear USDC and inverse BTC/ETH-settled options, typically targeting roughly **10-21 days to expiration (DTE)**, with filters on delta, OTM, liquidity, spread ratio, APR, and margin. A regime framework (normal / elevated / crisis) pauses new risk in crisis; circuit breakers include hard stops, soft triggers (roll preferred), take-profit, and time-based exits. **Naked short** is introduced below on this page with an illustrative payoff; **bull put spread** and **covered call** follow on the next pages. A comparison chart and summary table close the brief.

Strategy 1 of 3

Naked short

Single-leg premium collection; tail exposure is the highest of the three.



Figure 1. Illustrative naked short put expiry payoff (normalized units; not a forecast).

Mechanics

Sell one out-of-the-money (OTM) option per leg and collect premium. Configuration can restrict to short puts, short calls, or **both**; in **both** mode, put and call candidates compete under the same ranking key for available slots (no reserved allocation for calls).

Key risks

Tail risk is the highest among the three: a short put loses progressively as the underlying falls through the strike; a short call loses as price rallies. Extreme moves can stress margin and execution (spreads, slippage). Operational and model risk remain (scan logic, rolls, exits).

Return profile

Premium income dominates when short volatility and direction are favorable. Within this project narrative, expected yield is framed as **between covered call and bull put spread**.

Bull put spread

Credit spread with defined downside versus a naked short put.

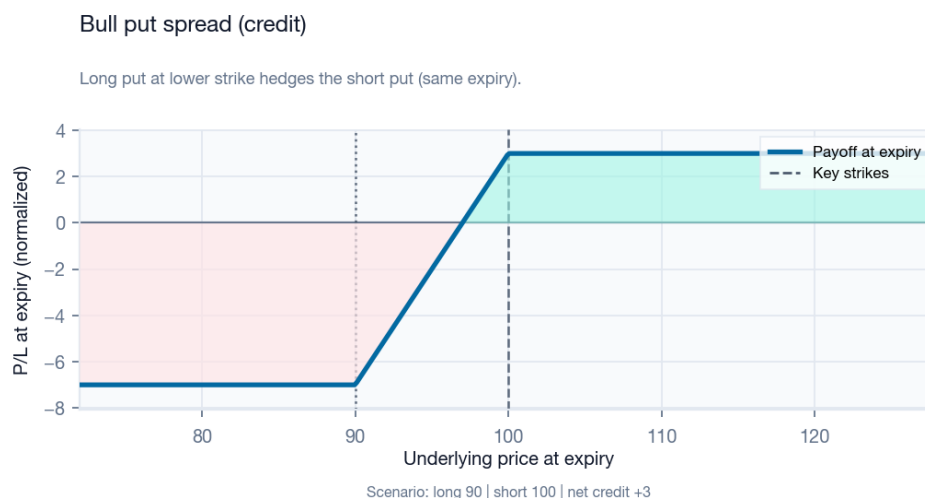


Figure 2. Illustrative bull put spread expiry payoff (normalized units; not a forecast).

Mechanics

Buy a lower-strike long put (protection) and sell a higher-strike short put for a net credit, same expiry. Maximum loss is conceptually capped at roughly the spread width minus the net credit received (fees and slippage apply). The long leg is selected within configured delta bounds and must sit below the short strike.

Key risks

Downside is **bounded** versus a naked short put, but the portfolio can still realize the full defined loss. Thin net credits amplify fee drag. Illiquid long legs can worsen effective protection and exit costs.

Return profile

Maximum profit per spread is typically the net credit if held to a benign outcome. In this project framing, **expected return is the most conservative** of the three, traded for capped loss.

Covered call

Spot inventory plus short call; Figure 3 shows coin-settled incremental BTC vs spot-only; Figure 4 shows linear USD terminal value of the covered package (capped at $K + \text{premium}$).



Figure 3. Covered call in **coin terms** (inverse / coin-settled): extra BTC at expiry vs holding spot only (illustrative; not a forecast).

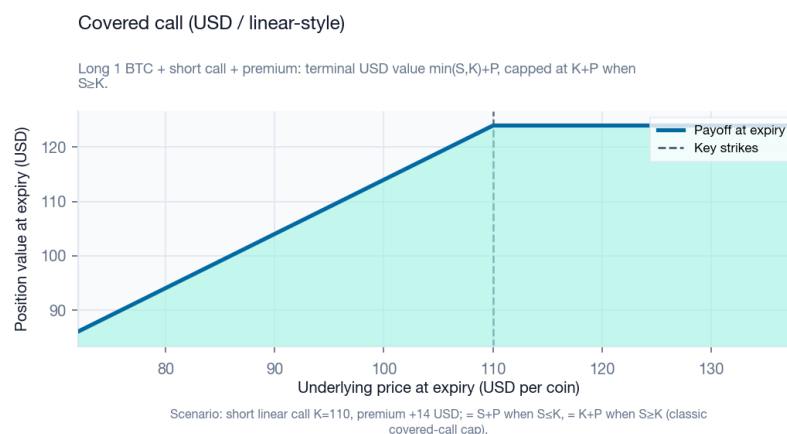


Figure 4. Covered call in **USD terms** (linear / USDC-settled): position value at expiry $\min(S, K) + \text{premium}$, flat above K so upside is **capped at $K + \text{premium}$** (illustrative; not a forecast).

Mechanics

Sell calls only when sufficient BTC or ETH inventory is already available in the account; the engine does not auto-buy spot for cover or use perpetuals as a substitute. Optional spot-exit flows can accompany in-the-money exits when enabled. On Deribit, **inverse** options settle in coin (BTC or ETH): premium is collected in coin, and when a short call expires in the money, intrinsic is settled as a coin outflow (conceptually *about* $\max(S-K, 0)/S$ coins per BTC of notional—Figure 3). **Linear USDC** legs settle intrinsic in USDC; for one BTC covered, terminal USD value is $S - \max(S-K, 0) + \text{premium} = \min(S, K) + \text{premium}$, so when $S \geq K$ the payoff plateaus at $K + \text{premium}$ (Figure 4—not a comparison to naked spot). The engine scans both quote styles; the two figures separate inverse vs linear accounting.

Key risks

Upside participation is **capped** near the short call strike after strong rallies. Spot inventory still bears downside price risk (premium provides partial cushion only). Settlement mechanics may leave residual spot exposure after cash or coin settlement.

Return profile

Combines spot exposure with call premium. In this project narrative, **expected return is the highest** among the three, alongside upside caps and inventory requirements.

Summary & comparison

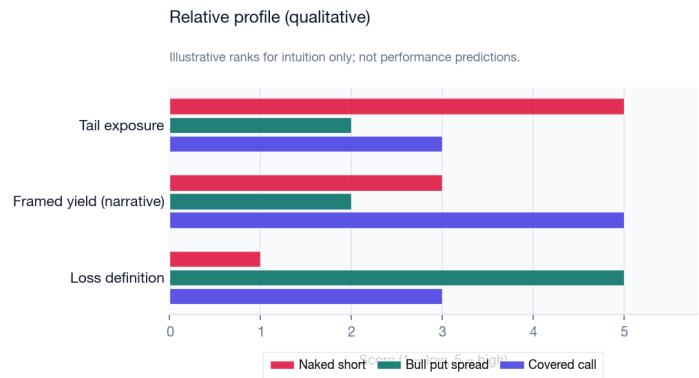


Figure 5. Relative scores are illustrative ranks for intuition; they are not performance predictions.

Use Figure 5 as a **mental map**: bull put spreads emphasize bounded loss; naked shorts emphasize premium at the cost of tail exposure; covered calls blend inventory economics with call premium but cap upside.

At a glance

Strategy	Core idea	Dominant risk	Return (project framing)
Naked short	Single-leg OTM short put/call/both	Tail / extreme moves	Mid vs. the other two
Bull put spread	Short put + lower long put	Defined loss to max width	Most conservative
Covered call	Spot + short call	Capped upside; spot drawdowns	Highest expected

Important limitations. Crypto derivatives involve substantial volatility and margin risk. Exchange, operational, and smart-routing failures can occur. Past or simulated performance does not guarantee future results. Prospective participants should use test environments and independent professional advice as appropriate.

This document describes strategy mechanics aligned with the engine repository design. It is not affiliated with or endorsed by Deribit.